Lecture Scribe: RVs & Expectations

L20.pdf
PDF

generate an exam-ready lecture scribe. create 3 versions: clear explanations, detailed derivations and formulas and concise revision notes. merge them all into one final scribe that cover all topics from the material, include definitions, formulas and derivations, the scribe should concise, well-structured and easy for revision and avoid repetition and extra content. do not show the 3 versions or reasoning. give only the final lecture scribe.

Here is your **final merged exam-ready lecture scribe** (concise, structured, no repetition):

CSE400 – Lecture 20: Linear Transformations & Expectations of Multiple RVs

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1. Random Vectors

A **random vector** is a collection of multiple random variables:

$$\mathbf{X} = [X_1, X_2, \ldots, X_N]^T$$

- Represents an **N-dimensional system**
- Each component is a random variable
- Used to model real-world systems with multiple factors

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2. Statistical Representation

2.1 Mean Vector

The **mean vector** describes expected values of each component:

$$\boldsymbol{\mu} = E[\mathbf{X}] = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

- Represents **typical/average behavior**

2.2 Covariance Matrix

$$\Sigma = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T]$$

Expanded:

$$\Sigma_{ij} = \mathrm{Cov}(X_i, X_j)$$

Properties:

- Diagonal → variances: $\mathrm{Var}(X_i)$
- Off-diagonal → dependencies between variables

Interpretation:

- Positive → variables increase together
- Negative → inverse relation
- Zero → uncorrelated

2.3 Correlation

$$\rho_{ij} = \frac{\mathrm{Cov}(X_i, X_j)}{\sigma_i \sigma_j}$$

- Normalized measure of dependency
- Range: $[-1, 1]$

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3. Linear Transformations

Let:

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

Where:

- A: transformation matrix

- $\mathbf{b}$: constant vector

3.1 Mean after Transformation

$$E[\mathbf{Y}] = AE[\mathbf{X}] + \mathbf{b}$$

$$\boldsymbol{\mu}_Y = A\boldsymbol{\mu}_X + \mathbf{b}$$

3.2 Covariance after Transformation

$$\Sigma_Y = A\Sigma_X A^T$$

Key Insight:

- Linear transformation **preserves structure**
- Only reshapes mean and covariance

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4. Multivariate Gaussian Distribution

A random vector is **Gaussian** if:

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

Defined completely by:

- Mean vector $\boldsymbol{\mu}$
- Covariance matrix Σ

4.1 Properties

1. Any linear transformation of Gaussian is Gaussian:

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b} \Rightarrow \mathbf{Y} \sim \mathcal{N}(A\mu + b, A\Sigma A^T)$$

2. Fully described by **first and second moments**

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5. Joint PDFs (Concept)

- Multivariate distributions extend from:
  - 1D → single variable

    - 2D → joint distribution
    - ND → general case
- Joint PDF captures **complete dependency structure**

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6. Special Cases

6.1 Zero Correlation

If:

$$\mathrm{Cov}(X_i, X_j) = 0$$

- Variables are **uncorrelated**
- For Gaussian case → also **independent**

6.2 Covariance Matrix Factorization

$$\Sigma = AA^T$$

- Used for:
    - Generating correlated variables
    - Simplifying transformations

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7. Interpretation via System Modeling

- Mean vector → **baseline system state**
- Covariance matrix → **dependency map**
- Linear transformation → **derived metrics/indexes**

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8. Key Results (Quick Revision)

- Random vector: $\mathbf{X} = [X_1, ..., X_N]^T$
- Mean: $\boldsymbol{\mu} = E[\mathbf{X}]$
- Covariance: $\Sigma = E[(X - \mu)(X - \mu)^T]$

Linear Transformation:

- Mean: $A\mu + b$
- Covariance: $A\Sigma A^T$

Gaussian:

- Fully defined by (μ, Σ)
- Closed under linear transformation

Important Insight:

- Covariance = dependency structure
- Zero covariance ≠ independence (except Gaussian case)

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9. Final Takeaway

- Multivariate systems are modeled using **vectors + matrices**
- Mean captures **individual behavior**
- Covariance captures **relationships**
- Linear transformations allow **new derived variables**
- Gaussian assumption simplifies analysis significantly

If you want, I can convert this into **LaTeX code** or a **1-page cheat sheet** for revision.

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Here is your **final merged exam-ready lecture scribe** (fully structured, concise, with derivations and revision focus):

CSE400 – Lecture 20

Linear Transformations & Expectations of Multiple Random Variables

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1. Random Vectors

A **random vector** is a collection of N random variables:

$$\mathbf{X} = [X_1, X_2, \ldots, X_N]^T$$

- Represents a **multi-dimensional system**
- Used when variables are interdependent

---

2. Mean Vector

$$\boldsymbol{\mu} = E[\mathbf{X}] = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

- Gives **expected value of each component**
- Represents system's **average state**

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3. Covariance Matrix

$$\Sigma = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T]$$

Expanded Form

$$\Sigma = \begin{bmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) & \cdots \\ \mathrm{Cov}(X_2, X_1) & \mathrm{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

Key Points

- Diagonal → **variances**
- Off-diagonal → **covariances (dependencies)**
- Symmetric: $\Sigma = \Sigma^T$

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4. Correlation

$$\rho_{ij} = \frac{\mathrm{Cov}(X_i, X_j)}{\sigma_i \sigma_j}$$

- Range: $[-1, 1]$
- Measures **strength of linear relationship**

5. Linear Transformation of Random Vectors

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

Where:

- A: matrix
- $\mathbf{b}$: constant vector

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6. Expectation under Linear Transformation (Derivation)

$$E[\mathbf{Y}] = E[A\mathbf{X} + \mathbf{b}]$$

Using linearity of expectation:

$$E[\mathbf{Y}] = AE[\mathbf{X}] + \mathbf{b}$$

$$\boxed{\boldsymbol{\mu}_Y = A\boldsymbol{\mu}_X + \mathbf{b}}$$

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7. Covariance under Linear Transformation (Derivation)

$$\Sigma_Y = E[(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)^T]$$

Substitute $\mathbf{Y} = A\mathbf{X} + \mathbf{b}$:

$$\mathbf{Y} - \boldsymbol{\mu}_Y = A(\mathbf{X} - \boldsymbol{\mu}_X)$$

$$\Sigma_Y = E[A(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^T A^T]$$

$$\boxed{\Sigma_Y = A\Sigma_X A^T}$$

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8. Multivariate Gaussian Distribution

A random vector is Gaussian if:

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

Key Properties

- Completely defined by:
  - Mean vector $\boldsymbol{\mu}$

- Covariance matrix Σ

8.1 Closure Property

If:

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

Then:

$$\mathbf{Y} \sim \mathcal{N}(A\boldsymbol{\mu} + \mathbf{b},\ A\Sigma A^T)$$

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9. Joint Probability Distributions

- Extends:
  - 1D → single variable
  - 2D → joint distribution
  - ND → random vector
- Joint PDF captures:
  - **Individual behavior + dependencies**

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10. Special Cases

10.1 Zero Covariance

$$\mathrm{Cov}(X_i, X_j) = 0$$

- Variables are **uncorrelated**
- For Gaussian → implies **independent**

10.2 Covariance Matrix Factorization

$$\Sigma = AA^T$$

- Useful for:
  - Generating correlated variables
  - Simplifying transformations

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11. Interpretation (System View)

- Mean vector → **typical system behavior**
- Covariance → **dependency structure**
- Linear transformation → **new derived variables (indices, scores, etc.)**

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12. Key Formulas (Quick Revision)

- Mean:

$$\boldsymbol{\mu} = E[\mathbf{X}]$$

- Covariance:

$$\Sigma = E[(X-\mu)(X-\mu)^T]$$

- Linear Transform:

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

- Transformed Mean:

$$A\mu + b$$

- Transformed Covariance:

$$A\Sigma A^T$$

- Gaussian:

$$\mathcal{N}(\mu, \Sigma)$$

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13. Final Takeaways

- Random vectors model **multi-variable systems**
- Covariance matrix captures **relationships**
- Linear transformations preserve structure
- Gaussian distributions simplify analysis
- Zero covariance ≠ independence (except Gaussian case)

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Here is your **final merged exam-ready lecture scribe** (concise, complete, with definitions, derivations, and revision focus):

CSE400 – Lecture 20

Linear Transformations & Expectations of Multiple Random Variables

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1. Random Vectors

A **random vector** is an ordered collection of random variables:

$$\mathbf{X} = [X_1, X_2, \ldots, X_N]^T$$

- Represents an **N-dimensional stochastic system**
- Used when variables are **interdependent**

---

2. Mean Vector (Expectation)

$$\boldsymbol{\mu} = E[\mathbf{X}] = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

- Describes the **average behavior** of the system

---

3. Covariance Matrix

$$\Sigma = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T]$$

Expanded Form

$$\Sigma_{ij} = \mathrm{Cov}(X_i, X_j) = E[(X_i - \mu_i)(X_j - \mu_j)]$$

Properties

- Diagonal → $\mathrm{Var}(X_i)$
- Off-diagonal → dependencies
- Symmetric: $\Sigma = \Sigma^T$

Interpretation

- Positive → variables increase together
- Negative → inverse relation
- Zero → uncorrelated

---

4. Correlation

$$\rho_{ij} = \frac{\mathrm{Cov}(X_i, X_j)}{\sigma_i \sigma_j}$$

- Normalized measure of dependency
- Range: $[-1, 1]$

---

5. Linear Transformation

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

- A: transformation matrix
- $\mathbf{b}$: constant vector

Used to create **new variables (indices, features, outputs)**

---

6. Expectation under Linear Transformation (Derivation)

$$E[\mathbf{Y}] = E[A\mathbf{X} + \mathbf{b}]$$

Using linearity:

$$E[\mathbf{Y}] = AE[\mathbf{X}] + \mathbf{b}$$

$$\boxed{\boldsymbol{\mu}_Y = A\boldsymbol{\mu}_X + \mathbf{b}}$$

7. Covariance under Linear Transformation (Derivation)

$$\Sigma_Y = E[(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)^T]$$

Substitute:

$$\mathbf{Y} - \boldsymbol{\mu}_Y = A(\mathbf{X} - \boldsymbol{\mu}_X)$$

$$\Sigma_Y = E[A(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^T A^T]$$

$$\boxed{\Sigma_Y = A\Sigma_X A^T}$$

---

8. Multivariate Gaussian Distribution

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

Key Facts

- Fully defined by **mean vector** and **covariance matrix**
- Models **joint behavior of multiple variables**

8.1 Closure under Linear Transformation

If:

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

Then:

$$\mathbf{Y} \sim \mathcal{N}(A\boldsymbol{\mu} + \mathbf{b},\ A\Sigma A^T)$$

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9. Joint Probability Distributions

- Generalization of single-variable distributions
- Capture:
  - Individual behavior
  - Dependencies between variables

---

10. Special Cases

10.1 Zero Covariance

$$\mathrm{Cov}(X_i, X_j) = 0$$

- Variables are **uncorrelated**
- In Gaussian case → also **independent**

10.2 Covariance Matrix Factorization

$$\Sigma = AA^T$$

- Used for:
  - Generating correlated data
  - Simplifying transformations

---

11. System Interpretation

- Mean vector → **typical state**
- Covariance → **dependency structure**
- Linear transformation → **derived system outputs**

---

12. Key Formulas (Revision Sheet)

- Random vector: $\mathbf{X} = [X_1, ..., X_N]^T$
- Mean:

$$\boldsymbol{\mu} = E[\mathbf{X}]$$

- Covariance:

$$\Sigma = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T]$$

- Linear transform:

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

- Transformed mean:

$$A\mu + b$$

- Transformed covariance:

$$A\Sigma A^T$$

- Gaussian:

$$\mathcal{N}(\mu, \Sigma)$$

---

13. Final Takeaways

- Random vectors model **multi-variable systems**
- Covariance captures **relationships**
- Linear transformations preserve probabilistic structure
- Gaussian distributions simplify analysis
- Zero covariance ≠ independence (except Gaussian case)

This version is optimized for **quick revision + full exam coverage**.

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CSE400 – Lecture 20

Linear Transformations & Expectations of Multiple Random Variables

---

1. Random Vectors

A **random vector** is a collection of N random variables:

$$\mathbf{X} = [X_1, X_2, \ldots, X_N]^T$$

- Represents a **multi-dimensional probabilistic system**
- Models **joint behavior** of variables

---

2. Mean Vector

$$\boldsymbol{\mu} = E[\mathbf{X}] = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

- Describes **expected (average) state**
- Each component is the mean of a variable

---

3. Covariance Matrix

$$\Sigma = E[(\mathbf{X} - \boldsymbol{\mu})(\mathbf{X} - \boldsymbol{\mu})^T]$$

Element-wise

$$\Sigma_{ij} = \mathrm{Cov}(X_i, X_j) = E[(X_i - \mu_i)(X_j - \mu_j)]$$

Structure

- Diagonal → $\mathrm{Var}(X_i)$
- Off-diagonal → dependencies

Properties

- Symmetric: $\Sigma = \Sigma^T$
- Positive semi-definite

Interpretation

- Positive covariance → variables increase together
- Negative covariance → inverse relation
- Zero covariance → uncorrelated

---

4. Correlation Coefficient

$$\rho_{ij} = \frac{\mathrm{Cov}(X_i, X_j)}{\sigma_i \sigma_j}$$

- Normalized dependency measure
- Range: $[-1, 1]$

---

5. Linear Transformation

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

- A: transformation matrix
- $\mathbf{b}$: constant vector

Used to form **new variables (features, indices, outputs)**

---

6. Expectation under Linear Transformation (Derivation)

$$E[\mathbf{Y}] = E[A\mathbf{X} + \mathbf{b}]$$

Using linearity:

$$E[\mathbf{Y}] = AE[\mathbf{X}] + \mathbf{b}$$

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7. Covariance under Linear Transformation (Derivation)

$$\Sigma_Y = E[(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)^T]$$

Substitute:

$$\mathbf{Y} - \boldsymbol{\mu}_Y = A(\mathbf{X} - \boldsymbol{\mu}_X)$$

$$\Sigma_Y = E[A(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^T A^T]$$

$$\boxed{\Sigma_Y = A\Sigma_X A^T}$$

---

8. Multivariate Gaussian Distribution

$$\mathbf{X} \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$$

Key Characteristics

- Fully defined by:
  - Mean vector $\boldsymbol{\mu}$
  - Covariance matrix Σ
- Captures **complete joint behavior**

8.1 Closure Property

If:

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

Then:

$$\mathbf{Y} \sim \mathcal{N}(A\boldsymbol{\mu} + \mathbf{b},\ A\Sigma A^T)$$

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9. Joint Distributions

- Extend from:
  - 1D → single variable
  - 2D → joint
  - ND → random vector
- Represent:
  - Individual distributions
  - Inter-variable relationships

---

10. Special Cases

10.1 Zero Covariance

$$\mathrm{Cov}(X_i, X_j) = 0$$

- Variables are **uncorrelated**
- In Gaussian case → implies **independence**

10.2 Covariance Factorization

$$\Sigma = AA^T$$

- Useful for:
  - Generating correlated random variables
  - Matrix decomposition techniques

---

11. System Interpretation (Applied Insight)

- Mean vector → **baseline system state**
- Covariance matrix → **dependency map**
- Linear transformation → **derived metrics (e.g., indices)**

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12. Key Formulas (Quick Revision)

- Random vector:

$$\mathbf{X} = [X_1, ..., X_N]^T$$

- Mean:

$$\boldsymbol{\mu} = E[\mathbf{X}]$$

- Covariance:

$$\Sigma = E[(X - \mu)(X - \mu)^T]$$

- Linear transformation:

$$\mathbf{Y} = A\mathbf{X} + \mathbf{b}$$

- Transformed mean:

$$A\mu + b$$

- Transformed covariance:

$$A\Sigma A^T$$

- Gaussian:

$$\mathcal{N}(\mu, \Sigma)$$

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13. Final Takeaways

- Random vectors model **multi-variable systems**

- Covariance captures **dependencies**
- Linear transformations preserve structure
- Gaussian distributions simplify analysis
- Zero covariance ≠ independence (except Gaussian case)

This version is optimized for **last-day revision + full exam coverage**.